

Polysaccharides Isolated from Açáí Fruit Induce Innate Immune Responses

The Açáí (Acai) fruit is a popular nutritional supplement that purportedly enhances immune system function. These anecdotal claims are supported by limited studies describing immune responses to the Acai polyphenol fraction. Previously, we characterized $\gamma\delta$ T cell responses to both polyphenol and polysaccharide fractions from several plant-derived nutritional supplements. Similar polyphenol and polysaccharide fractions are found in Acai fruit. Thus, we hypothesized that one or both of these fractions could activate $\gamma\delta$ T cells. Contrary to previous reports, we did not identify agonist activity in the polyphenol fraction; however, the Acai polysaccharide fraction induced robust $\gamma\delta$ T cell stimulatory activity in human, mouse, and bovine PBMC cultures. To characterize the immune response to Acai polysaccharides, we fractionated the crude polysaccharide preparation and tested these fractions for activity in human PBMC cultures. The largest Acai polysaccharides were the most active in vitro as indicated by activation of myeloid and $\gamma\delta$ T cells. When delivered in vivo, Acai polysaccharide induced myeloid cell recruitment and IL-12 production. These results define innate immune responses induced by the polysaccharide component of Acai and have implications for the treatment of asthma and infectious disease.